EXPERIMENT 20: Prolog Program for PLANETS DB

Aim

To write and execute a Prolog program to create a database of planets and retrieve information about planets using suitable queries.

Algorithm

1. Start the Prolog program.
2. Define facts containing information about planets.
3. Store the planet information in the knowledge base.
4. Define queries to retrieve required planet information.
5. Execute the queries.
6. Display the matching planet details.
7. Stop.

Program Code:

```
% Planet database


planet(mercury, terrestrial).
planet(venus, terrestrial).
planet(earth, terrestrial).
planet(mars, terrestrial).
planet(jupiter, gas_giant).
planet(saturn, gas_giant).
planet(uranus, ice_giant).
planet(neptune, ice_giant).


% Check planet type
planet_type(Planet, Type) :-
    planet(Planet, Type).
```

Output:

```
yes
| ?- consult('C:/Users/Hp/Documents/ex-20.pl').
compiling C:/Users/Hp/Documents/ex-20.pl for byte code...
C:/Users/Hp/Documents/ex-20.pl compiled, 13 lines read - 1142 bytes written, 16 ms

yes
| ?- planet_type(earth, Type).

Type = terrestrial

yes
| ?- planet_type(Planet, gas_giant).

Planet = jupiter ?
```

Result

The Prolog program for the Planet database was successfully implemented. Planet information was stored as facts and retrieved correctly using Prolog queries.